Project 22: Movie Revenue Prediction

Preliminary Report

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1 Task Description

The primary goal of this project is to create a machine learning model capable of predicting movie revenue. Because the target variable—revenue—is a continuous numerical value, we are framing this task as a **regression** problem. The model will take a set of specific movie attributes as inputs and output a continuous revenue prediction.

2 Dataset Description

The data for this project is sourced from Kaggle. The dataset consists of various movie attributes that will serve as our input features, such as the movie's budget, release date, and original language. To prepare the data for modeling, we will perform extensive preprocessing, which will include cleaning the data, handling missing values, analyzing feature distributions, and splitting the data into appropriate training and testing sets. We will also explore the possibility of data augmentation or sourcing additional data to improve model robustness.

3 Proposed Solution

For structured, tabular regression problems like box office prediction, the usual methods include Linear Regression, Support Vector Regression (SVR), Random Forests, and Gradient Boosting Machines. We plan to implement and compare the following three methods to find the optimal architecture for our computational resources and final performance:

- **Ridge Regression (Baseline):** A regularized linear regression model. We will use this as our foundational baseline model to establish an initial performance metric. The L2 regularization helps prevent overfitting, which can occur when dealing with numerous movie attributes.
- **Random Forest Regressor:** An ensemble learning method that constructs a multitude of decision trees during training and outputs the average prediction of the individual trees. This method is highly effective for tabular data, handles non-linear relationships well, and is generally robust to outliers in features like budget or revenue.
- **XGBoost Regressor (Extreme Gradient Boosting):** A highly efficient and scalable implementation of gradient boosted decision trees. XGBoost builds trees sequentially, where each new tree attempts to correct the residual errors of the previous ones. It frequently achieves state-of-the-art performance on Kaggle tabular datasets and will serve as our primary advanced model.